Connecting Parameter Uncertainties to Model Uncertainties for Tropical Precipitation Projections

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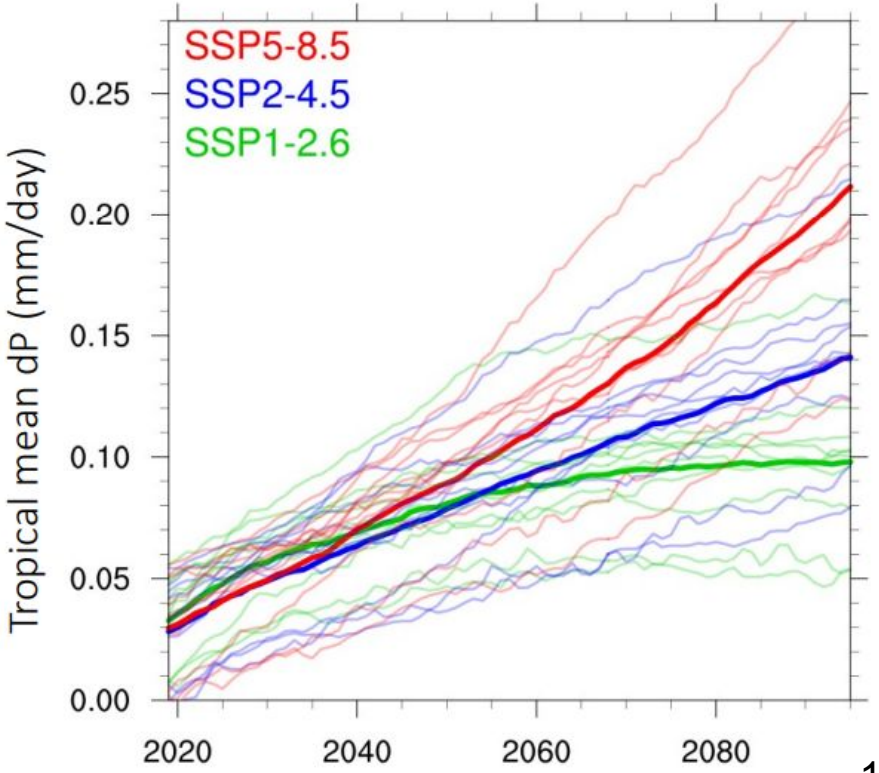
July 17, 2023

What are climate models?

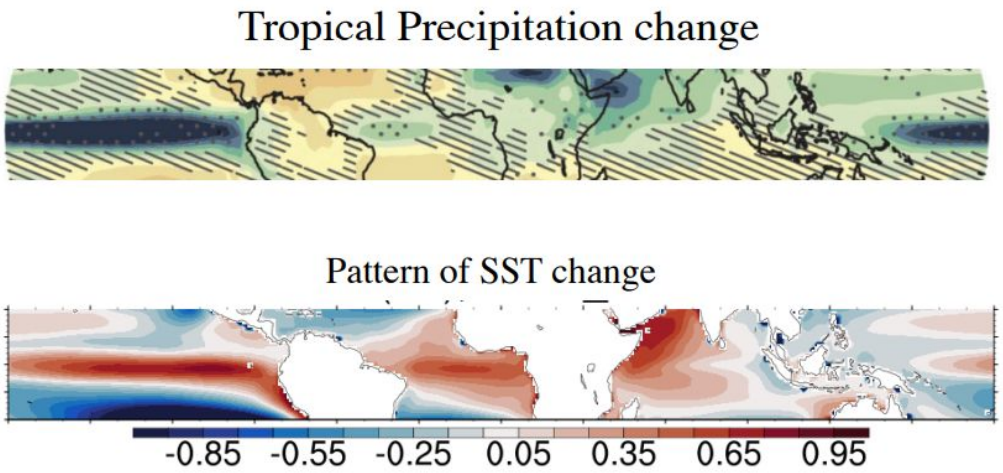
Climate models predict future environmental conditions

- Uncertainty exists in model precipitation projections



How do precipitation projections work?

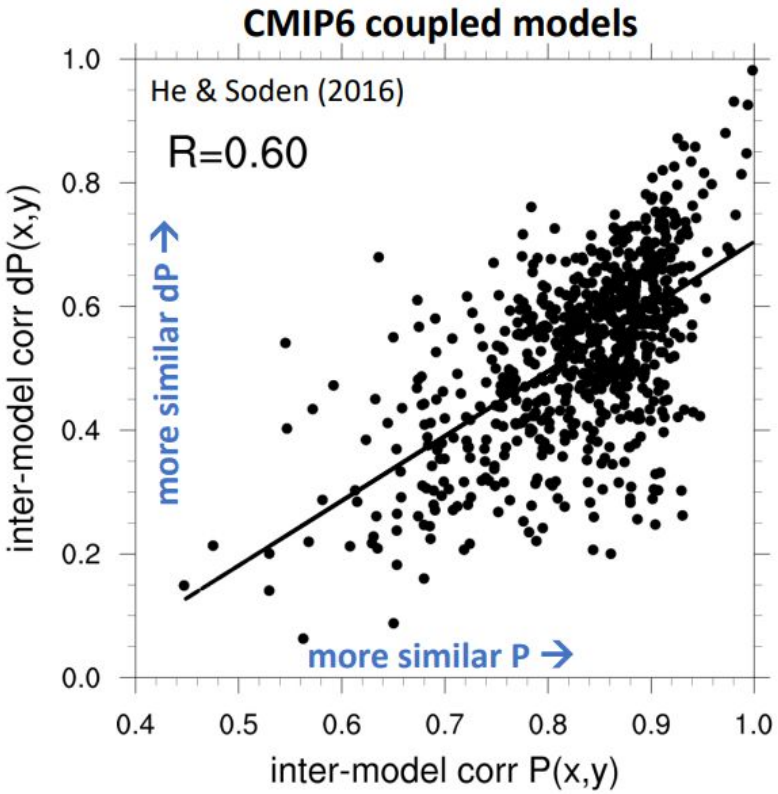
- In global climate models a connection exists between present day sea surface temperature (SST) and precipitation



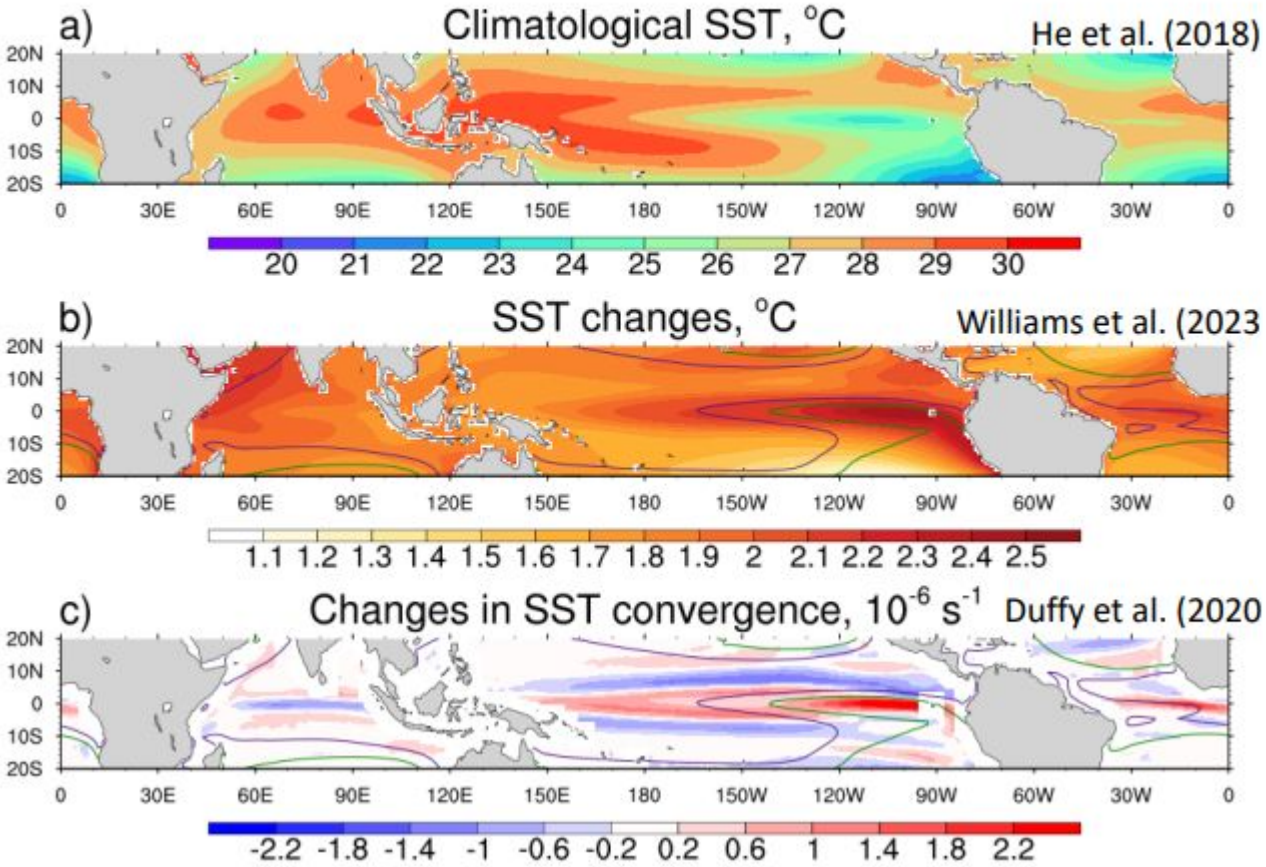
**This is referred to as warmer-get-wetter*

What are climate models?

- Models with similar precipitation climatology project similar regional precipitation changes



How do precipitation projections work?



What is a two-mode model?

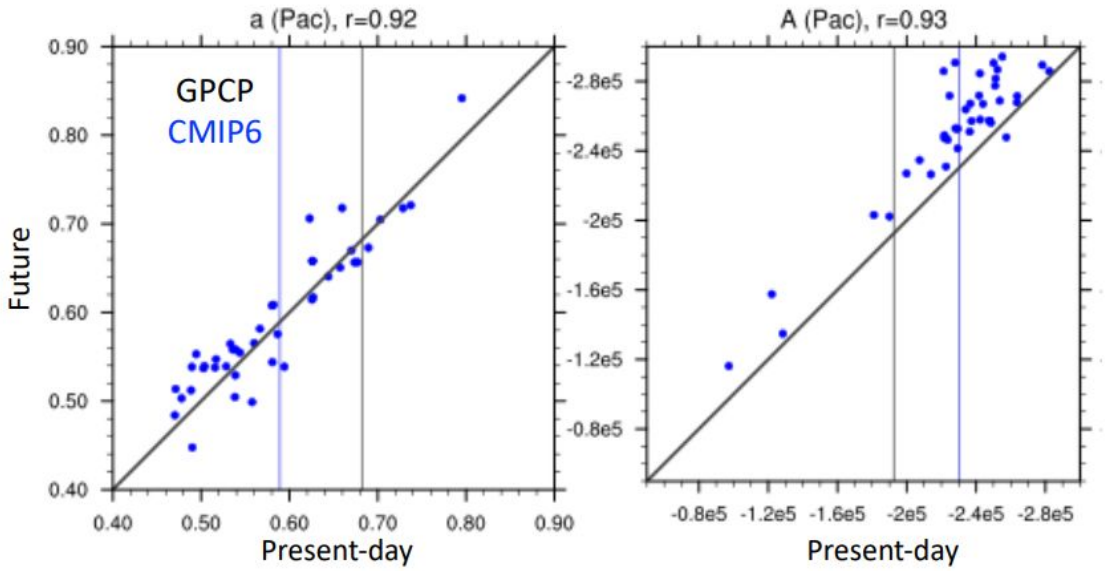
- Precipitation projections are driven by SST and surface convergence (SC) of winds

$$P^*(SST_{rel}) + P^*(SC) = C \cdot exp(a \cdot SST) + A \cdot SC + R$$

- Where a is an unknown relating to SST
- Where A is an unknown relating to SC

What is a two-mode model?

- Present day P/SST relationship can be used to constrain dP/dSST and dP/dSC
- Models underestimate a (SST term) and overestimate A (SC term)



What does this mean for model constraints?

- There is a lack of constraints on regional precipitation change in the tropics
- Present-day precipitation could be a constraint on future precipitation change

How can we improve precipitation projections?

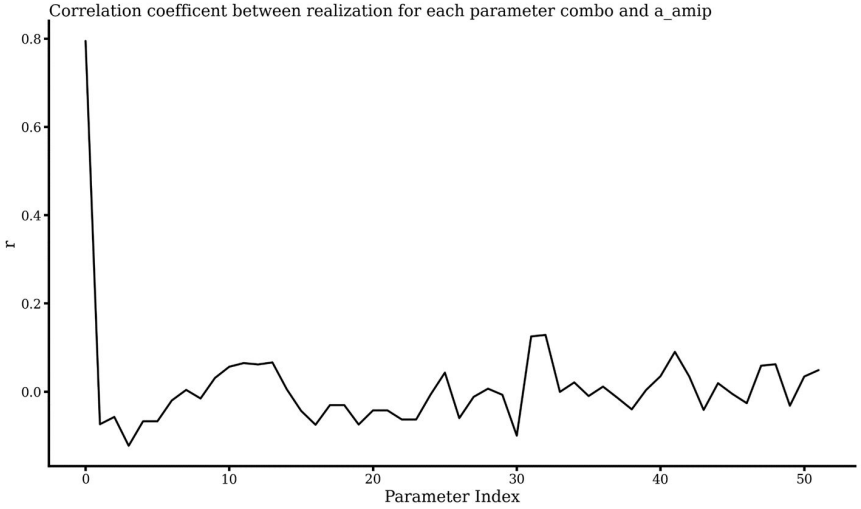
What is a perturbed physics ensemble?

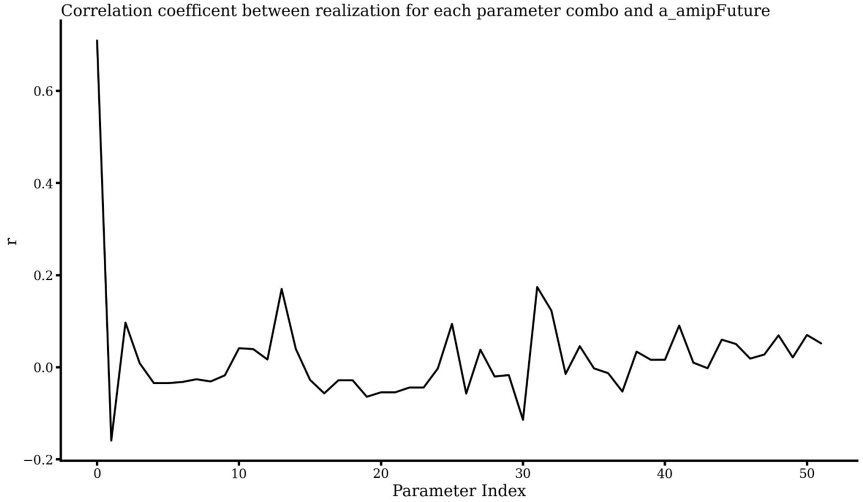
- Involves parameters being adjusted to see a wide range of future climate conditions

- Helps to identify what parameter is most responsible for uncertainties in projections

What parameters impact precipitation projections?

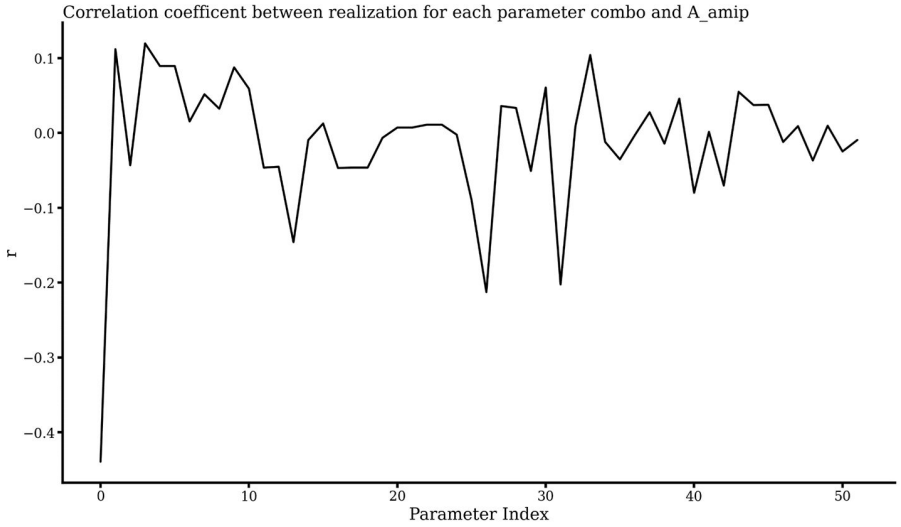
- Parameter 0, deep entrainment rate, has the greatest correlation between the constants a and A

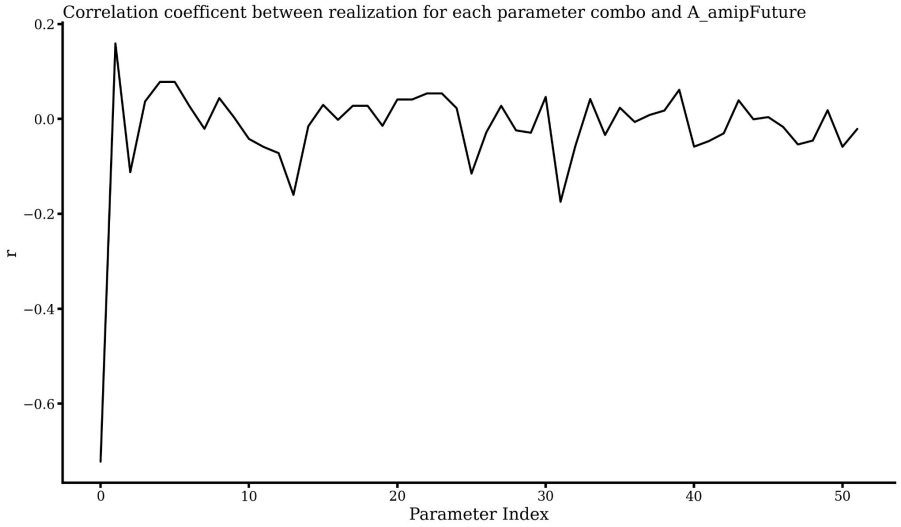




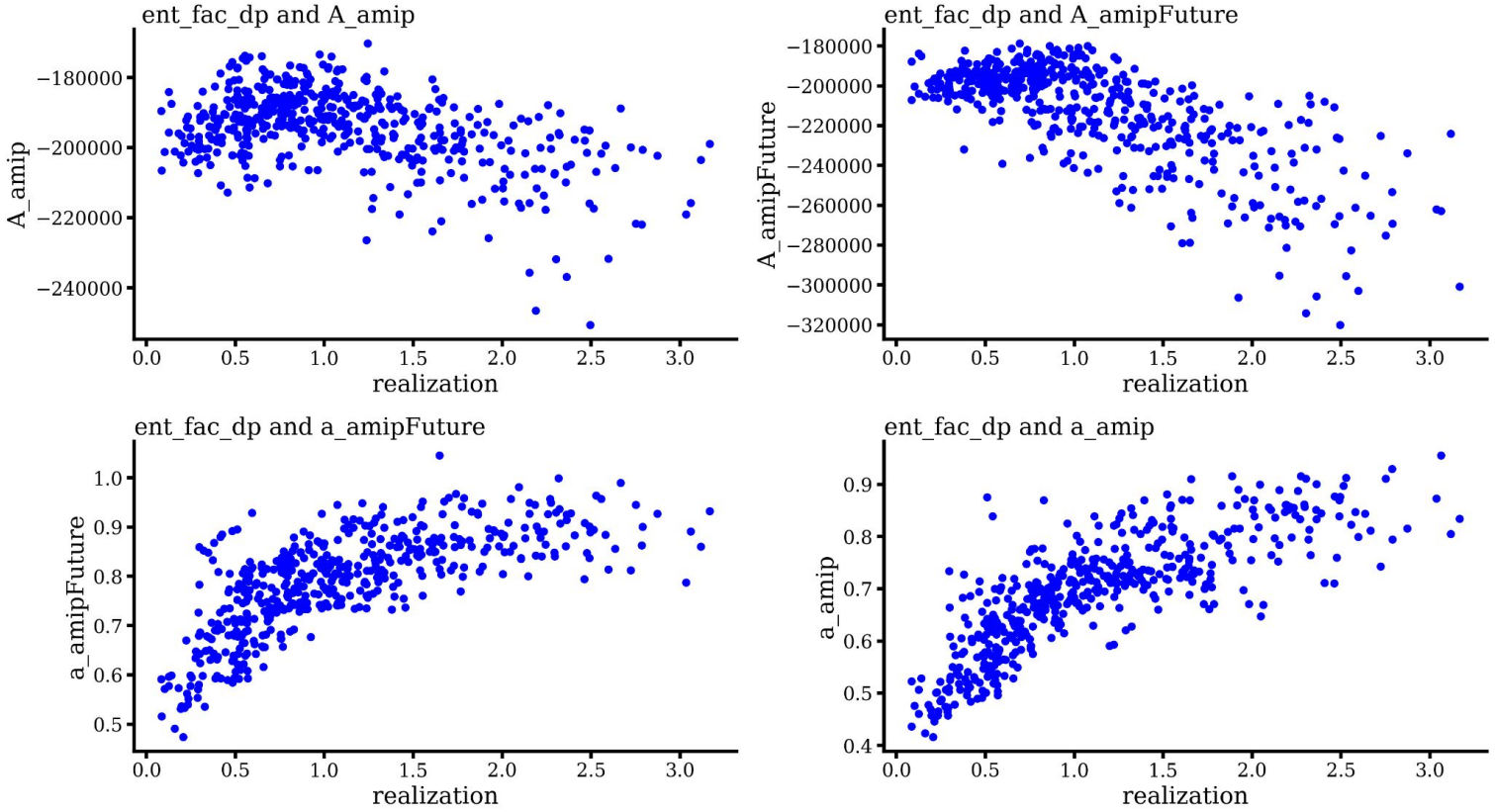
What parameters impact precipitation projections?

- Deep entrainment rate has largest impact on precipitation projections





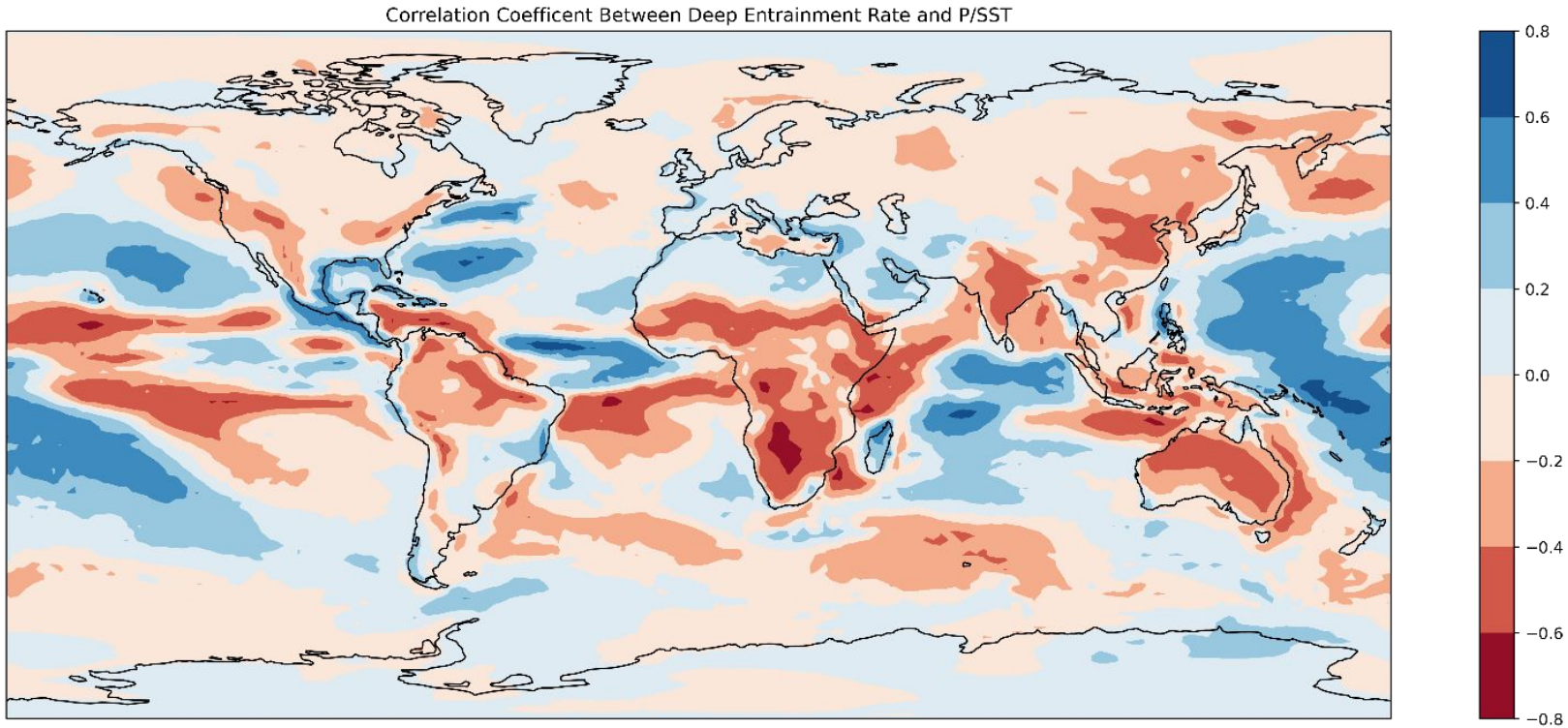
Closer look at deep entrainment rate



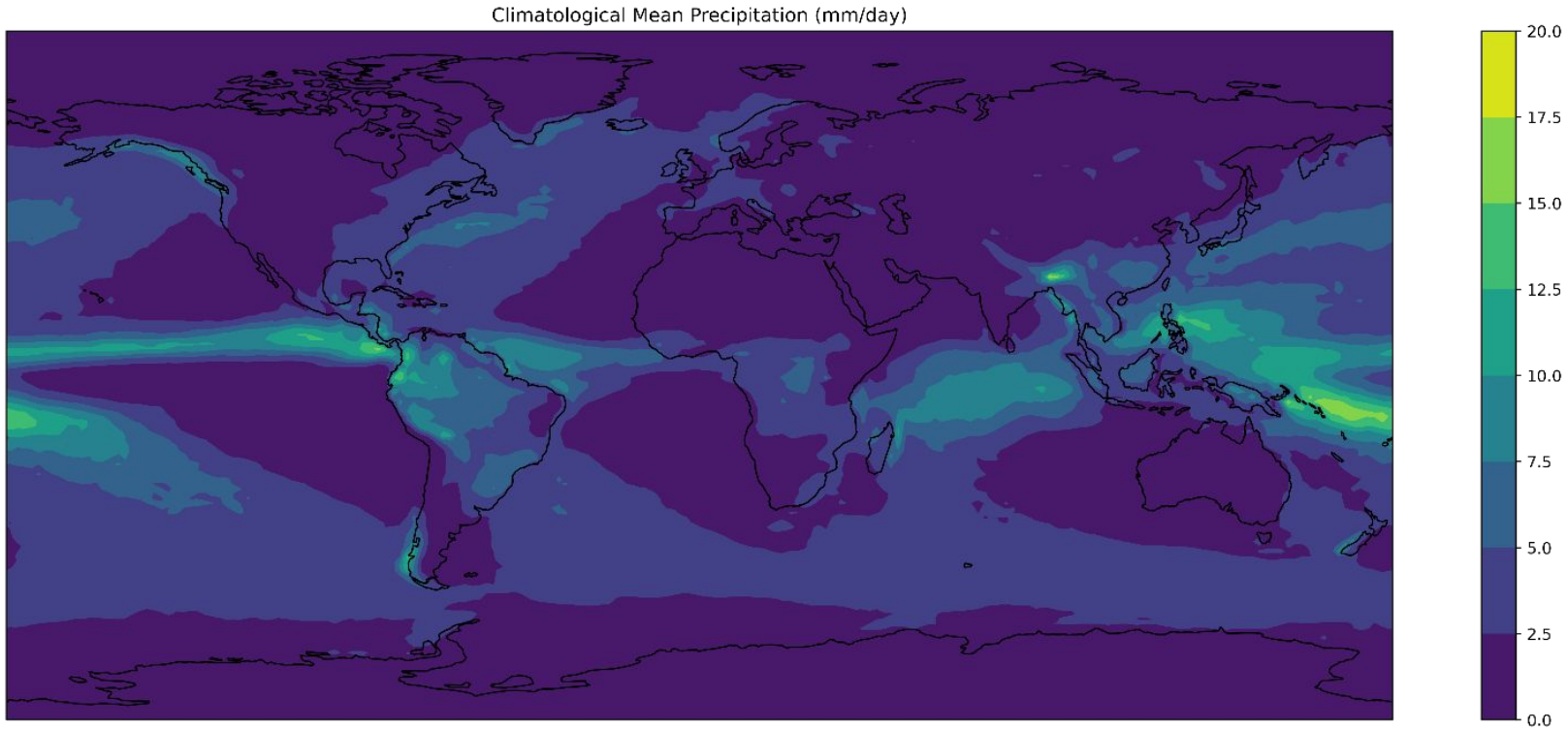
Deep entrainment rate

- Parcel of air from environment is pulled into cloud formation
- Cloud will take on properties of that air parcel
- Overtime, this leads to the air taking on the qualities of the parcel

Deep entrainment rate correlation



Deep entrainment rate correlation



Conclusions

Tuning deep entrainment rate can reduce uncertainty in tropical precipitation projections

Present-day precipitation could be a constraint on future precipitation change

Acknowledgments

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Acknowledgments

Thanks to National Science Foundation 1852040

Thank you for your attention

Any questions?